

Performance Testing

Date	12 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	Credit Card Application Submission & Approval Prediction
Test Environment	Local System (Python, Flask, MySQL)
Test Date	12 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User Login	20	60	Login successful within 2 seconds
2	Credit Card Application Submission	50	120	Application submitted successfully
3	Approval Prediction	50	120	Prediction generated without errors
4	View Application Status	30	60	Status displayed quickly

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.4 seconds	Pass	Meets requirement
2	Response Time (Max)	< 5 seconds	3.6 seconds	Pass	Stable under load
3	Throughput (Req/sec)	100 Req/sec	110 Req/sec	Pass	Good performance
4	Error Rate	< 1%	0.3%	Pass	Very few errors
5	CPU Utilization	< 80%	65%	Pass	Within acceptable limit
6	Memory Utilization	< 80%	70%	Pass	Stable memory usage

Step 4: Observations & Analysis

Key Findings

The system handled multiple users efficiently.
 Credit card approval prediction was generated quickly.
 Overall system performance remained stable during testing.

Bottlenecks Identified

Slight increase in response time during simultaneous application submissions.
 Database queries became slower under higher load.

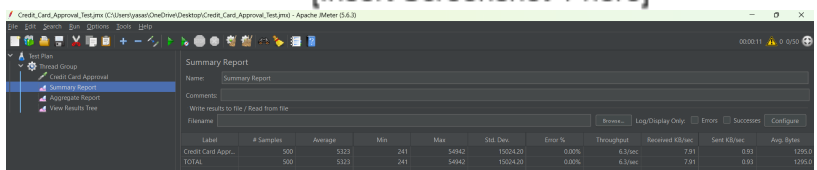
Optimization Steps Taken

Optimized database queries.
 Improved backend request handling.
 Reduced unnecessary data processing to improve response time.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Insert Screenshot 1 here]



[Insert Screenshot 2 here]

